PICOPATT Internship

Progress Review — April 2026

Explainable Clustering for Spatio-Temporal Picoclimatic Data

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Current Research Objectives

Foundation

- Keep explanations in raw sensor units: temperature, humidity, wind speed.
- Map indices like s_1 or t_{41} back to physical traits.
- Measure stability with Jaccard overlap of top SHAP features across 6 seeds.
- Expand beyond KMeans / ExKMC to autoencoder and spatial baselines.

Delivery Target

1–2 manuscripts

by early August

methods + application

Internship Arc: March → August 2026

Lit Review Output

- 3-axis taxonomy (representation / clustering / explanation)
- Novel 4-term joint objective proposed
- Review draft complete

Implementation Sprint Output

- Full pipeline running on 4 proxy datasets
- SHAP, Abductive, and ExKMC benchmarked
- Explanation stability via top-5 SHAP overlap

Current Pipeline Architecture

DATA
$\sim$23 vars $\times$ 4 timesteps/day (PICOPATT sensor data)

↓

REPRESENTATION
Raw time series | Shapelet distances (mixed-length: 10–30, adaptive dictionary)

↓

CLUSTERING
KMeans | ExKMC (interpretable tree output)

↓

EXPLANATION
SHAP (feature attribution) | Abductive (minimal edits) Surrogate fidelity: train 0.955 / CV 0.93

↓

PRESENTATION
Sensor-readable cluster summary $\rightarrow$ urban planners/architects [format to be frozen]

From whiteboard $\rightarrow$ now fully implemented over a 2-week sprint.

Key Results from the Implementation Sprint

0.95/0.93

Surrogate fidelity (train / CV)

Defends explanation faithfulness

+17.6%

Silhouette gain mixed vs fixed shapelets

Consistent across all 3 datasets

6 seeds

Explanation stability quantified

Top-5 feature overlap 67–100% across datasets

3 ablation datasets

Cross-dataset ablation table

ECG200, ECG5000, Roma taxi

Mixed vs Fixed Shapelet Length — Cross-Dataset Ablation

Dataset	Fixed sil.	Mixed sil.	Gain
ECG200 (96 pts)	0.5697	0.6503	+14.1%
ECG5000 (140 time points)	0.5920	0.6845	+15.6%
Roma taxi (8 engineered features)	0.5545	0.6820	+23.0%
Average			**+17.6%**

Current Status: Validated vs Still Needed

✓ Validated

- End-to-end pipeline on 4 proxy datasets
- KMeans + ExKMC both running & comparable
- SHAP + Abductive explanations implemented
- Surrogate fidelity: train 0.955 / CV 0.93
- Mixed-length shapelets outperform fixed (+17.6%)
- Explanation stability: 6 seeds, Jaccard top-feature overlap metric
- Synthetic picoclimatic data generator validated
- HDBSCAN spatial baseline added
- Autoencoder + KMeans baseline confirmed runnable

! Still Needed

- Real PICOPATT sensor data (not yet available)
- Explanation in raw sensor units (degrees Celsius, %, m/s) — not s_1/t_{41}
- Explanation output format frozen as contract
- Deep method: autoencoder pipeline with richer XAI
- Spatial density baseline for picoclimatic windows
- Extended seed sweep ($\geq$10 seeds) for paper claims
- Paper direction confirmed: methods vs application

The Core Research Problem

An urban planner selects a spatial region and time window. The system clusters the pico-climatic windows. The explanation says: **"Cluster 2 = high humidity variance + low wind speed in building canyons."** This narrative is identical whether they pick a 1-hour or 6-hour window.

Stable Format

Explanation must not change when window size or feature format changes.

Sensor Language

$s_1 \rightarrow$ "Humidity Variance"

$t_{41} \rightarrow$ "Temperature at 10 PM"

Alignment Gap

Sequential pipelines break alignment. Joint optimisation is the target.

Proposed Directions — Phase 2

Direction A — Neural Representation

- Autoencoder $\rightarrow$ latent space $\rightarrow$ KMeans
- TELL-style joint neural clustering
- Interpretability gap: explain $\phi(X)$ vs X
- **Paper: methods paper on XAI pipeline**

Higher complexity · CNN vision

Direction B — Statistical Aggregation

- Mean, Var, Quartiles per variable (Time2Feat)
- Directly in sensor units $\rightarrow$ no mapping needed
- Clean baseline for stable explanation format
- **Paper: application paper on PICOPATT**

Simpler · natural first comparator

Proposal (Hybrid): B as the clean sensor-unit baseline $\rightarrow$ A as the comparator. *Shared comparison target: performance vs interpretability, with one frozen explanation format.*

Decision Points for Today's Meeting

01 Real PICOPATT data?

A pivot to real sensor data should be made only after the proxy benchmark is complete. Premature transition would reduce methodological generality.

02 Paper angle?

A methods paper (XAI pipeline + shapelet ablation) or an application paper (PICOPATT urban-planning results) should be selected for the August timeline.

03 Freeze the format?

The sensor-readable output schema should be locked before more methods are added. Design choices are constrained, but format drift is prevented.

04 ExKMC or XClusters?

ExKMC should remain the primary explainable baseline, or XClusters can be prototyped (co-adaptive clustering + tree). XClusters is more novel; ExKMC is already working.

Proposed Timeline to August 2026

W1–2 Apr 25–May 8	W3–4 May 9–22	W5–6 May 23–Jun 5	W7–8 Jun 6–19	W9–10 Jun 20–Jul 3	W11–15 Jul–Aug
Real picoclimate data Freeze explanation format Add GMM + spatial baseline validation	Statistical aggregation (B) Time2Feat sensor-unit features Stability across methods	Autoencoder + KMeans (A) Interpretability gap analysis Latent vs raw unit comparison	Joint objective prototype (ExKMC + spatial regulariser) Cross-method XAI table	Paper draft: methods + results Figures finalised Supervisor review	Revision, feedback Internship report Paper submission

Scope Reminder

Current data status

The real PICOPATT data is still unavailable for the current context. The picoclimate work in the repository is a simulated test fixture used for validation only.

- Simulated picoclimate fixture: 4 timesteps, about 23 variables, used to test the pipeline structure.
- Real PICOPATT validation remains future work once the real data is available.
- Current focus: keep the explanation format stable and broaden the benchmark matrix.

What Changed Recently

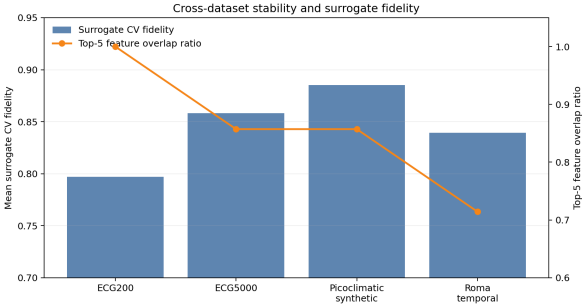
Validated in code

- Shapelet notebooks renamed and aligned under `shapelets_`
- Cross-dataset stability run completed with 6 seeds
- Raw-vs-shapelet sanity check completed: mixed results by dataset
- HDBSCAN added to the picoclimate benchmark script
- Autoencoder + KMeans baseline rerun: silhouette 0.218 $\rightarrow$ 0.360 (test_deep)
- Runtime audit added under `outputs/timings`: stability 14.3 min (quick) vs 119.3 min (full)

Governance update

- Explanation output format is now treated as a contract
- Future methods must emit the same cluster-summary structure
- The goal is sensor-readable explanations, not feature-index-only narratives

Evidence: Stability Across Datasets

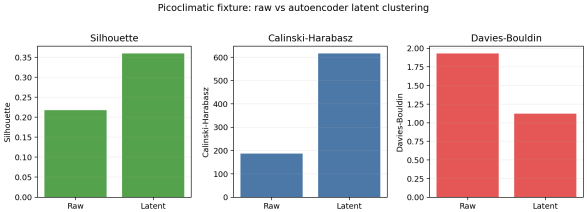


What this slide should show

- Surrogate CV fidelity band across the 6-seed sweep
- Top-5 SHAP overlap across 6 seeds
- Runtime context: full stability sweep took about 119 minutes in the dated rerun
- The main robustness evidence

- Reserve this space for the stability summary figure when you want the slide to show the actual chart.

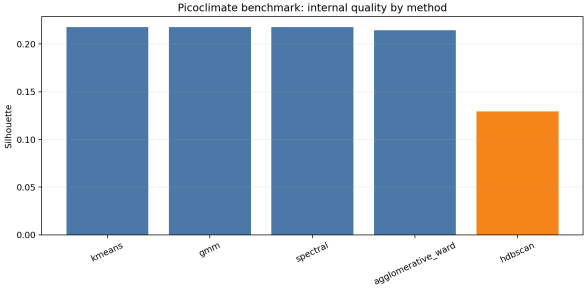
Evidence: Deep Representation Baseline



Interpretation

- Autoencoder baseline is the deep comparator
- Latest test_deep rerun improves silhouette from 0.218 to 0.360
- Useful for the interpretability-gap story

Evidence: Spatial Baseline Expansion



Interpretation

- HDBSCAN broadens the baseline set beyond KMeans / ExKMC
- Latest test_benchmark run still favors KMeans (silhouette 0.218, k=2)
- HDBSCAN remains useful as the density/noise comparator in the matrix
- Final PICOPATT benchmark still pending real data

Interpretation Layer

What is already supported

- Surrogate fidelity is high enough to defend model-faithful explanations.
- SHAP concentration already gives a compact narrative for the shapelet baseline.
- Stability is now measured rather than assumed.
- Shapelets help on some datasets, but the raw-vs-shapelet result is dataset-dependent.

What is still missing

- Sensor-unit explanation template frozen across methods
- Spatial density comparators on real PICOPATT data
- A fuller joint-optimization method if the paper goes beyond a pipeline story

Recommended Next Phase

1. Freeze the explanation output format and make every method emit the same cluster-summary schema.
2. Keep ExKMC as the main interpretable clustering baseline for now.
3. Extend the benchmark matrix with spatial methods; HDBSCAN is the first one now in place.
4. Treat the autoencoder + KMeans path as the deep representation baseline.
5. If the research contribution shifts toward methods, prototype the simplified ExKMC-style spatial regularizer next.

Practical takeaway

The current repository is now ready for a Phase 2 story: simulated validation, stable explanations, and broader clustering baselines before moving to real PICOPATT data.